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Key Points:
- Two critical regions responsible for modified Circumpolar Deep Water intrusions are identified in Prydz Bay
- Both the eastern coastal current and the Prydz Bay gyre serve to transport modified Circumpolar Deep Water to the ice calving front
- The need to reproduce dynamic effects of small-scale troughs on the Antarctic Slope Current is highlighted

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Citation:
Liu, C., Z. Wang, C. Cheng, R. Xia, B. Li, and Z. Xie (2017), Modeling modified Circumpolar Deep Water intrusions onto the Prydz Bay continental shelf, East Antarctica, *J. Geophys. Res. Oceans*, *122*, doi:10.1002/2016JC012336.

Received 13 SEP 2016
Accepted 7 MAY 2017
Accepted article online 15 MAY 2017

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Modeling modified Circumpolar Deep Water intrusions onto the Prydz Bay continental shelf, East Antarctica

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Abstract An eddy-resolving coupled regional ocean-sea ice-ice shelf model is employed to locate the hot spots where modified Circumpolar Deep Water (mCDW) intrudes onto the continental shelf within Prydz Bay, and locate the paths through which mCDW is transported to the Amery Ice Shelf (AIS) calving front. Evaluation of the model output is with satellite, hydrographic and borehole data. Two critical windows responsible for mCDW intrusions are identified. The first is the eastern branch of the cyclonic Prydz Bay gyre (PBG) that carries mCDW to the ice front line, accounting for an annual mean heat transport of $\sim 8.7 \times 10^{11}$ J s^{-1}. The second is located to the east of the Four Ladies Bank (FLB) where mCDW is channeled through submarine troughs, accounting for an annual mean heat transport of $\sim 16.2 \times 10^{11}$ J s^{-1}. The eddy-induced heat transport accounts for ~23% in the path of the PBG and ~52% in the path of the eastern coastal current, with respect to their total onshore heat transport. The seasonal pulsing of mCDW intrusions is greatly dependent on the seasonal cycle of the Antarctic Slope Current (ASC) that peaks with a maximum of ~29.3 Sv at 75°E in June. In austral winter, mCDW is allowed to access the eastern flank of the AIS calving front with potential consequences for the basal mass balance of the AIS. The dynamic effects of small-scale troughs on the longshore ASC play an important role in the onshore mCDW transport.

1. Introduction

After the Weddell and Ross Seas, Prydz Bay is the third largest bay around the Antarctic and contains the third largest Antarctic ice shelf, the Amery Ice Shelf (AIS). Exchanges of mass, heat, salt, and nutrients occur between Prydz Bay and the Southern Ocean [*Heil et al.*, 1996; *Wong and Riser*, 2013]. Thus, climate change in the Southern Ocean [*Wang et al.*, 2015] has potential for significant impacts on the AIS [*Williams et al.*, 2002]. On the other hand, changes in the AIS thickness and extent could also potentially influence sea level change [*Shepherd et al.*, 2010]. Prydz Bay is host to a marine ecosystem that interacts with the physical environment. Understanding oceanographic processes within the bay and the sub-ice-shelf environment allows better interpretation of the nature of biological communities [*Williams et al.*, 2010; *Post et al.*, 2014].

Along with the seasonally varying sea ice, the ocean and ice-shelf components in Prydz Bay form a complex coupled system. However, a limited number of field studies provide some hydrographic observations for Prydz Bay, and these are largely confined to summer months. Based on geopotential anomalies derived from hydrographic profiles in austral summer, the large-scale circulation in Prydz Bay was described to be primarily composed of a cyclonic gyre, the Prydz Bay gyre (PBG), to the north of the AIS [*Smith et al.*, 1984; *Nunes Vaz and Lennon*, 1996]. The inflow of this gyre hugs the eastern rim of the bay, and favors the onshore intrusions of warmer modified Circumpolar Deep Water (mCDW) across the continental shelf break [*Heil et al.*, 1996]. Characterized as a salty, warm water mass approaching the shelf break, Circumpolar Deep Water (CDW) mixes with cold Shelf Water (SW) over the continental shelf and is transformed into denser mCDW [*Yabuki et al.*, 2006]. The onshore intrusions of mCDW could reach the AIS calving front and even the AIS cavity, to substantially enhance the basal melt rate of the AIS [*Herraiz-Borreguero et al.*, 2016].

As the largest embayed ice shelf in East Antarctica, the AIS is fed by the Lamber-Mellor-Fisher tributary glacial systems [*Wen et al.*, 2007; *Yu et al.*, 2010], with an area of ~62,000 km^2. The basal mass balance of the AIS reflects the ocean thermal forcing at the ice shelf-ocean interface, which is primarily dominated by the

circulation pattern and properties of Ice Shelf Water (ISW) in the sub-ice-shelf cavity [*Herraiz-Borreguero et al*., 2015]. Driven by the "ice-pump," the circulation beneath the AIS is generally steered by the shape of the sub-ice-shelf cavity [*Lewis*, 1985], suggesting that ocean circulation beneath the AIS and associated basal mass balance are very sensitive to the geometry of the cavity [*Hellmer and Jacobs*, 1992; *Galton-Fenzi et al*., 2008]. However, the thickness of the ice draft in contact with the ocean could be up to ~2500 m [*Fricker et al*., 2000], indicating difficulty in the in situ hydrography observations beneath the AIS. The limited observation and the difficulty in sampling the AIS and Prydz Bay have resulted in the use of numerical circulation models to develop process understanding of ocean and ice-shelf dynamics in this region [e.g., *Hellmer and Jacobs*, 1992; *Williams et al*., 2001, 2002; *Hemer et al*., 2006; *Galton-Fenzi et al*., 2008, 2012]. These numerical modeling studies of oceanographic processes under the AIS provide insights into the oceanic processes in the sub-ice-shelf cavity. They all reproduce a cyclonic circulation under the AIS, i.e., inflow of SW in the eastern flank and outflow of ISW in the western flank. The patterns of modeled ocean inflow and outflow in the sub-ice-shelf cavity are consistent with the composition and age of the surface sedimentary record collected from the AIS [*Post et al*., 2014], and the associated basal heat and freshwater fluxes are also reasonably characterized. However, comparing to observations from borehole sites, previous modeling studies have not captured mCDW intrusions across the AIS calving front, which substantially increases oceanic heat supply and consequently enhances the basal melt [*Herraiz-Borreguero et al*., 2015, 2016].

Indeed, seasonally varying intrusions of mCDW across the shelf break into Prydz Bay have already been revealed by analysis of the oceanic heat flux [*Heil et al*., 1996], with a stronger upwelling of warm deep water in winter [*Gordon*, 1981]. After mixing with cold SW over the continental shelf, CDW is transformed into mCDW that flows toward the AIS calving front by riding above the southward current. When approaching the AIS, mCDW would play an important role in the horizontal exchange of heat and salt across the ice-shelf front [*Williams et al*., 2001]. Thus, a better understanding of the process, by which the seasonally varying intrusions of mCDW influence the basal mass balance of the AIS, allows us to better interpret the seasonality of water mass properties, of circulations, and of the AIS basal mass balance in Prydz Bay.

In this study, we aim to document the location where mCDW is advected onto the continental shelf, and the paths through which mCDM is transported to the AIS calving front and into the sub-ice-shelf cavity. Furthermore, quantitative assessments of the transports of Antarctic Slope Current (ASC), water mass properties and the AIS melt rate are also described to aid our interpretation of the seasonal cycle of onshore flow and consequent basal melt rate. For these purposes, we set up an eddy-resolving regional coupled ocean-sea ice-ice shelf model. The main improvements of this coupled model over previous ones [*Williams et al*., 2001, 2002] are the use of a high accuracy estimate of the sub-ice-shelf-cavity geometry [*Timmermann et al*., 2010], an eddy-resolving spatial resolution, and the inclusion of sea-ice dynamics and thermodynamics. By increasing spatial resolution to adequately resolve meso-scale eddies, this model takes full advantage of the high-accuracy topography data set, which is particularly necessary for simulating the dynamic effects of small-scale troughs along the continental shelf break. The dynamic effects of troughs on the longshore ASC can create significant onshore flows, which hosts energetic meso-scale eddies induced by baroclinic instabilities [*Klinck and Dinniman*, 2010; *St-Laurent et al*., 2013; *Thompson et al*., 2014]. Thus, bathymetry needs to be represented at the meso-scale to capture the dynamic effects of topography on the ASC [*Klinck and Dinniman*, 2010; *Stewart and Thompson*, 2015]. In addition, the inclusion of an active sea ice component might be helpful for more realistic simulation of freshwater forcing associated with sea ice, a forcing that has often been estimated using surface heat flux.

The paper is organized as follows. Section 2 provides descriptions of the model and its implementation, and the hydrographic observations used for evaluating the simulations. Section 3 presents a detailed evaluation of the model simulations and model skill. The intrusions of mCDW and consequent impacts on the AIS are assessed in section 4. Section 5 summarizes and provides a discussion of the results.

2. An Eddy-Resolving Model and Hydrographic Data Sets

2.1. Model Description

The Massachusetts Institute of Technology General Circulation Model (MITgcm) is employed in this study [*Marshall et al*., 1997]. Based on the primitive equations with the Boussinesq approximation and hydrostatic

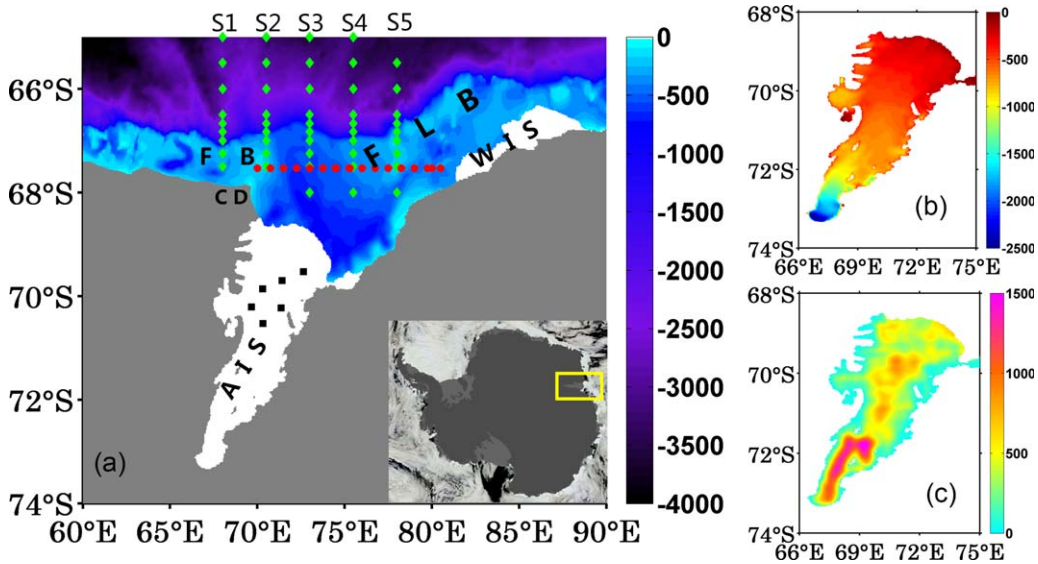

Figure 1. (a) Topographical features in Prydz Bay (m). AIS (Amery Ice Shelf), WIS (West Ice Shelf), PB (Prydz Bay), CD (Cape Darnley), FB (Fram Bank), FLB (Four Ladies Bank) are marked at the corresponding locations. (a) The yellow box in the inset shows the location of our region of interest. The sections of the 29th CHINARE voyage (green dots) shown in Figure 5 and Figure 6 are labeled by S1to S5. The section of JARPA voyage is indicated by red dots. The HWD of AMISOR shown in Figure 8 is marked as black squares. (b) The depths of the ice shelf draft (m). (c) The thickness of water column in the sub-ice-shelf cavity (m).

assumption, the MITgcm is coupled with a sea-ice model and an ice-shelf model. The inclusion of a dynamic and thermodynamic active sea-ice component is necessary for more realistic simulations of heat, salt and momentum exchanges among subcomponents [*Zhang and Hibler*, 1997; *Zhang and Rothrock*, 2000]. The ice-shelf component is used to represent a static and thermodynamically active ice shelf as the AIS in the south of the model domain [*Losch et al*., 2010].

Compared with previous modeling studies [*Williams et al*., 2001; *Galton-Fenzi et al*., 2012], our model domain covers a larger area, extending to an open northern boundary at 65°S, eastern boundary at 90°E, and western boundary at 60°E (Figure 1a). The horizontal grids are placed on a spherical orthogonal projection. The model resolution is 0.045° and 0.014° in the zonal and meridional directions, respectively. Accordingly, the zonal grid spacing ranges from ~1.4 km at the southern boundary (74°S) to ~2.2 km at the northern boundary (65°S), and there is a constant meridional grid spacing of ~1.5 km. As the Rossby radius of deformation is estimated at ~3 km over the continental shelf by the method in *Chelton et al*. [1998], this high horizontal resolution is necessary for the model to properly resolve meso-scale processes. Hence, there is no need to parameterize the eddy-induced transport, which can make comparable contributions to the Antarctic overturning circulation with the surface wind-driven transport [*Thompson et al*., 2014]. In the vertical direction, there are 70 levels, with intervals ranging from 10 m at the surface to 250 m at the bottom.

The ice draft and bathymetry are obtained from a global 1-min Refined Topography data set (RTopo-1) which contains self-consistent maps for upper and lower ice-surface heights, and seabed topography [*Timmermann et al*., 2010]. In RTopo-1, the high-resolution gridded data for cavity geometry shows that the deepest area of the AIS is up to ~2,500 m (Figure 1b), and the water column thickness in the sub-ice-shelf cavity has a maximum of > 1,500 m (Figure 1c). As suggested by *Galton-Fenzi et al*. [2012], these accurate representations of ocean-cavity geometry can greatly refine the simulated sub-ice-shelf circulation and melting/freezing rates of the AIS, which is a significant advance over previous modeling studies in Prydz Bay [*Williams et al*., 2001, 2002; *Hemer et al*., 2006].

The climatology of the Japanese Re-Analysis (JRA-25) at a spatial resolution of 1.125°×1.125° is used as the atmospheric boundary conditions, to force the ocean and sea ice components every 6 h, including downward longwave and shortwave radiation, precipitation, 2 m humidity, 2 m air temperature, 10 m surface wind, and river-runoff. The open eastern, western and northern boundary conditions are adapted from a cube sphere global experiment, the Estimating the Circulation and Climate of the Ocean phase II: high resolution global-ocean and sea-ice data synthesis (ECCO2) [*Menemenlis et al*., 2008]. As the spatial resolution of

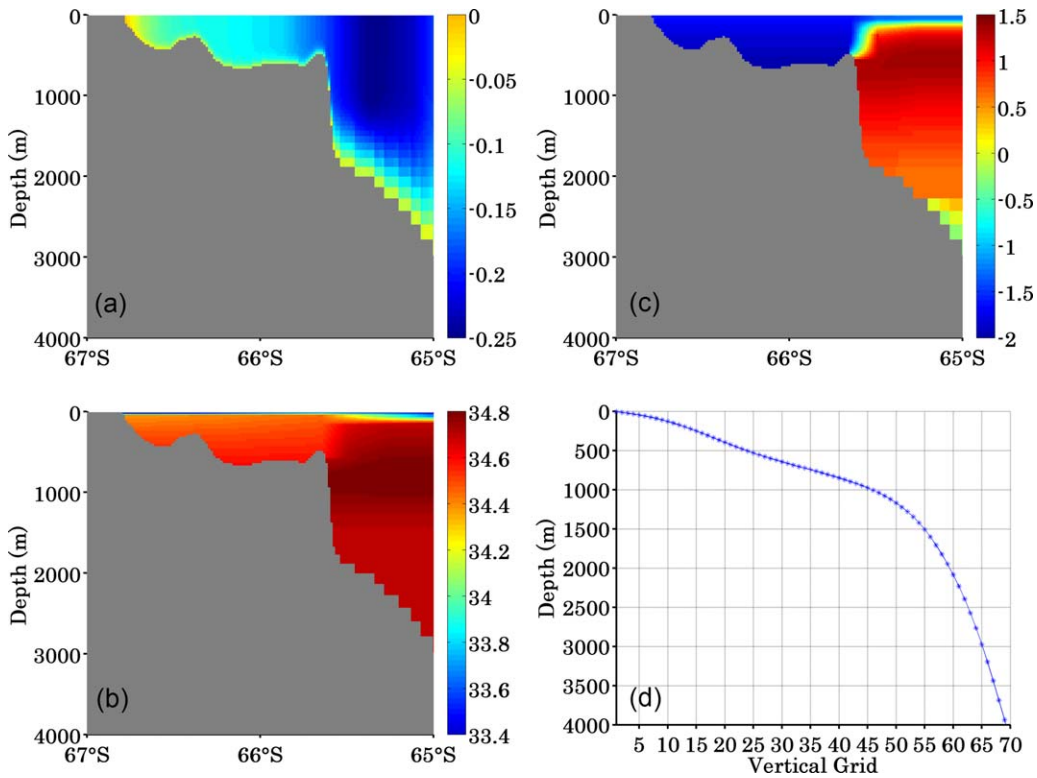

Figure 2. Vertical sections of the eastern open boundary conditions for: (a) zonal velocity in units of m s^{-1} (positive eastward/out of the page), (b) potential temperature in units of °C, (c) salinity in units of psu and (d) the distribution of the vertical levels. The grey region shows the sea bed.

our model is much higher than ECCO2 (the cube87 version of the CS510 free forward runs), the data near shelf regions need to be extrapolated for these open boundary conditions. In addition, the derivation of temperature and salinity for these open boundary conditions also considers the hydrographic data from a comprehensive interdisciplinary survey, the Baseline Research on Oceanography, Krill and the Environment [*Williams et al.*, 2010]. Based on observations of the ASC in *Heywood et al.* [1999], the zonal current at the eastern open boundary is modified to take into account the longshore ASC. Figure 2a shows that there is a strong westward jet along the shelf break as a representation of the ASC [see *Heywood et al.*, 1999, Figure 6b]. The potential temperature in shelf regions is set near the surface freezing point (−1.96°C), and cold SW is separated from warmer CDW by a marked thermocline, known as the Antarctic Slope Front (ASF) associated with the ASC (Figure 2b). In the deep ocean, CDW is represented by a warmer and saltier layer at the depths of 500–1000 m (Figures 2b and 2c).

The parameter settings are summarized as follows. Advection of temperature and salinity is performed by a second-order moment advection scheme [*Prather*, 1986]. A vertical Laplacian viscosity of 5×10^{-4} m^2 s^{-1} is applied for numerical stability. Strain rate-dependent diffusivity is used as the horizontal subgrid-scale viscosity and diffusion [*Smagorinsky*, 1963]. The background mixing of potential temperature and salinity is realized by a vertical diffusivity of 5×10^{-7} m^2 s^{-1}, supplemented by the K Profile Parameterization [*Large et al.*, 1994]. The quadratic bottom-drag coefficient at the sea bed is set as 2×10^{-3}. The oceanic equation of state is given by the numerical algorithm of *Jackett and McDougall* [1995].

The seawater properties in the sub-ice-shelf cavity need longer time to reach a steady state due to the slower hydrography adjustment beneath the ice shelf. In order to speed up the spin-up, the initial conditions of the seawater properties are derived from a previous version that has been integrated for 20 years, which shortens the spin up (see below). Thereafter, the model is integrated for 10 years again and saved with 1 day mean output. Note that our model has not included the frazil ice and tidal processes.

2.2. Hydrographic Data Sets

As the first coupled ocean-sea ice- ice shelf model for Prydz Bay, it is necessary to evaluate the model results by using comprehensive hydrographic data sets. For this purpose, five observation-based data sets are used in this study. The conductivity-temperature-depth (CTD) profiles obtained during the 29th Chinese Antarctic Expedition (CHINARE, the cruise from February 2013 to March 2013) are used to evaluate the model results in the deep ocean (Figure 1a). Hydrographic data from the Japanese Whale Research Program in the Antarctic (JARPA, the cruise from December to March during the austral summer during 1997–2004), which aims to improve the scientific information for the management of cetacean resources, is used to evaluate the simulated potential temperature and salinity over the continental shelf in summer. In this study, we choose one zonal expendable conductivity-temperature-depth (XCTD) profile from JARPA (Figure 1a). In the sub-ice-shelf cavity, hydrographic data are so sparse that the water mass features are poorly quantified based on purely *in situ* observations. So far, only data derived from Hot-Water Drilling (HWD) is available to examine and quantify the processes in the AIS cavity. The Amery Ice Shelf Ocean Research (AMISOR) project [*Craven et al*., 2004] drilled six boreholes (AM01-AM06) to access the sub-ice-shelf cavity (Figure 1a). Here, the salinity and potential temperature data from HWD at these sites is used to evaluate the simulated properties of ISW.

Marine animals equipped with CTD tags open a new approach to monitor oceanic processes. The instrumented southern elephant seal data, provided by the Integrated Marine Observing System, is an important source of hydrographic data during seasons with harsh weather around Antarctica [*Fedak*, 2004]. These instrumented southern elephant seals visit water columns randomly on the physical viewpoint; however, the seal data almost covers the entire shelf region of Prydz Bay over all the seasons. Thus, this approach offers a year-round hydrographic survey, which is particularly useful for evaluating the simulated mCDW on the continental shelf.

Supplied by the National Snow and Ice Data Center, the observed sea ice concentration data from Special Sensor Microwave Imager (SSM/I) is used to evaluate the simulated sea ice.

3. Evaluation of the Model Results

3.1. The Spin Up of the Model Simulation

In order to monitor if the model simulation approaches an equilibrium state, we calculate both the integrated Kinetic Energy (KE) and Heat Content (HC). The HC is defined as follows,

$$HC = \int T^K C_p \rho dV, \tag{1}$$

where T^k is the potential temperature in units of K, C_p is the specific heat capacity of 3974 J kg^{-1} K^{-1}, ρ is the seawater density (kg m^{-3}) and V is the volume of each cell (m^3). Considering that water masses in the sub-ice-shelf cavity adjust much more slowly than in the open sea, we also assess bulk measures for the sub-ice-shelf cavity separated from the entire domain.

The time series of integrated KE show that the model reaches an approximate steady state after integration of ~500 days (Figure 3a). The ice-pump [*Lewis*, 1985] is highly effective in driving strong circulations beneath the AIS, coinciding with modeling studies by *Schodlok et al*. [2016] using MITgcm (not shown). By contrast to the KE, it takes ~700 days to reach a steady state in the integrated HC, due to lesser sensitivity of tracers to boundary conditions (Figure 3b). When the result is nearly stable during the last 5 years, the average salinity in sub-ice-shelf cavity fluctuates near 34.43 psu, and the net basal melt rate fluctuates near ~1.6 m yr^{-1} (Figure 3c). The simulated net basal melt rate is relatively larger than previous modeling studies and observations [*Hellmer and Jacobs*, 1992; *Galton-Fenzi et al*., 2012; *Rignot et al*., 2013; *Depoorter et al*., 2013], and it might be due to the relatively thick vertical layers at the base of the ice shelf in our simulation [*Schodlok et al*., 2016]. As emphasized by *Schodlok et al*. [2016], the thickness of the water layers adjacent to the base of the ice shelf is important for basal melt rate estimates. In order to cool down the entire water layer with larger heat capacity, the water layer adjacent to the ice shelf base provides more heat to the ice shelf with thicker levels, thereby giving rise to a higher basal melting. Below, we use the last 5 years model output to conduct our analysis. In what follows, the austral summer is defined as the period from December to February in the next year, and the austral winter is June, July and August.

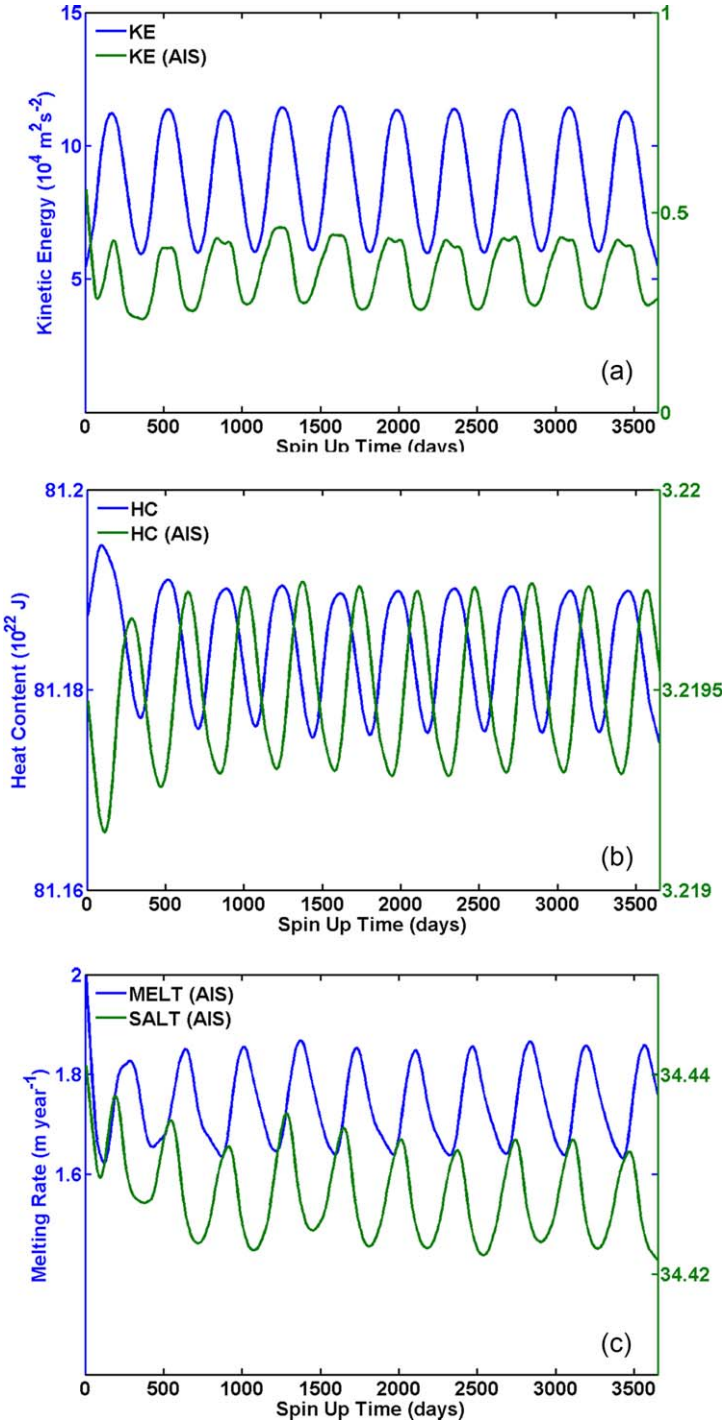

Figure 3. Time series of (a) the total Kinetic Energy (KE, in units of 10^4 m^2 s^{-2}) over the entire domain (blue line) and over the sub-ice-shelf cavity of the AIS (green line). (b) Same as Figure 3a, but for the sum of total Heat Content (HC, in units of 10^{22} J). (c) The average melting rate (blue line, in units of m year^{-1}) at the base of the AIS and the average salinity (green line, in units of psu) of water masses in the sub-ice-shelf cavity. 90 day moving averages were applied to the original time series.

3.2. Evaluation of the Simulated Sea Ice and Polynyas

In austral winter, the sea ice fully covers the entire Prydz Bay basin, except for a few areas, the polynyas, remaining open due to katabatic winds. The brine rejection within polynyas has potential for a significant impact on the production of denser water masses, e.g., the newly found source of AABW in Cape Darnley polynya (CDP) [*Ohshima et al.*, 2013; *Nakayama et al.*, 2014]. In previous modeling studies, sea-ice production was estimated by assuming that the surface heat loss during the freezing season is only compensated by new sea-ice formation [e.g., *Galton-Fenzi et al.*, 2012; *Nakayama et al.*, 2014]. In this study, we include more realistic sea-ice dynamic and thermodynamic processes to address seasonal sea-ice melting/freezing and transport.

A snap shot of the simulated sea-ice concentration in winter evidently shows three major polynyas in Prydz Bay (Figure 4a). The CDP is located to the west of the Fram Bank, the Mackenzie polynya (MP) is located to the north the AIS calving front, and the Prydz-Barrier polynya (PBP) occurs along the east coast of Prydz Bay. These simulated polynyas are consistent with previous studies in spatial distribution and temporal duration [*Tamura et al.*, 2008; *Kusahara et al.*, 2010; *Ohshima et al.*, 2013]. Furthermore, the significant seasonal cycle of sea ice is assessed by the total sea ice area, calculated by summing the sea ice area at each surface cell. The sea ice area essentially agrees with observations from the SSM/I (Figure 4b), with a high-coverage period from May to October and a minimum of 5×10^{10} m^2 in February. The larger area of sea-ice simulated than observed might result from the biases of the atmospheric forcing fields and the lack of frazil ice processes.

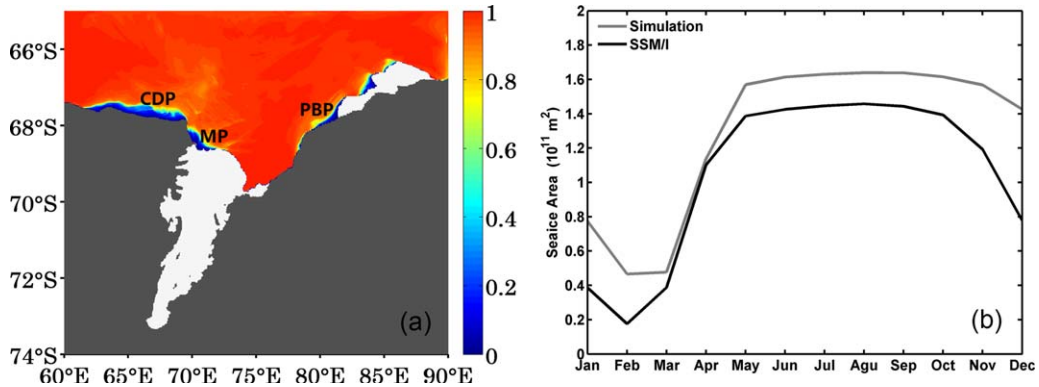

Figure 4. (a) A snapshot of the simulated sea ice concentration in August. Cape Darnley polynya (CDP), Mackenzie polynya (MP) and Prydz-Barrier polynya (PBP) are marked at their corresponding locations. (b) The simulated seasonal cycle of sea-ice-covered area (grey line) and satellite-derived observations (black line) in units of 10^{11} m^2.

3.3. Evaluation of Water Masses

Three primary types of water are identified in Figures 5 and 6, including Antarctic Surface Water (AASW), Winter Water (WW) and CDW. AASW is recognizable as a relatively warm and fresh layer above the seasonal thermocline, with its potential temperature warmer than 1°C due to summer heating (Figure 5) and salinity fresher than 33.6 psu due to sea ice melting (Figure 6). The thickness of AASW gradually increases to the

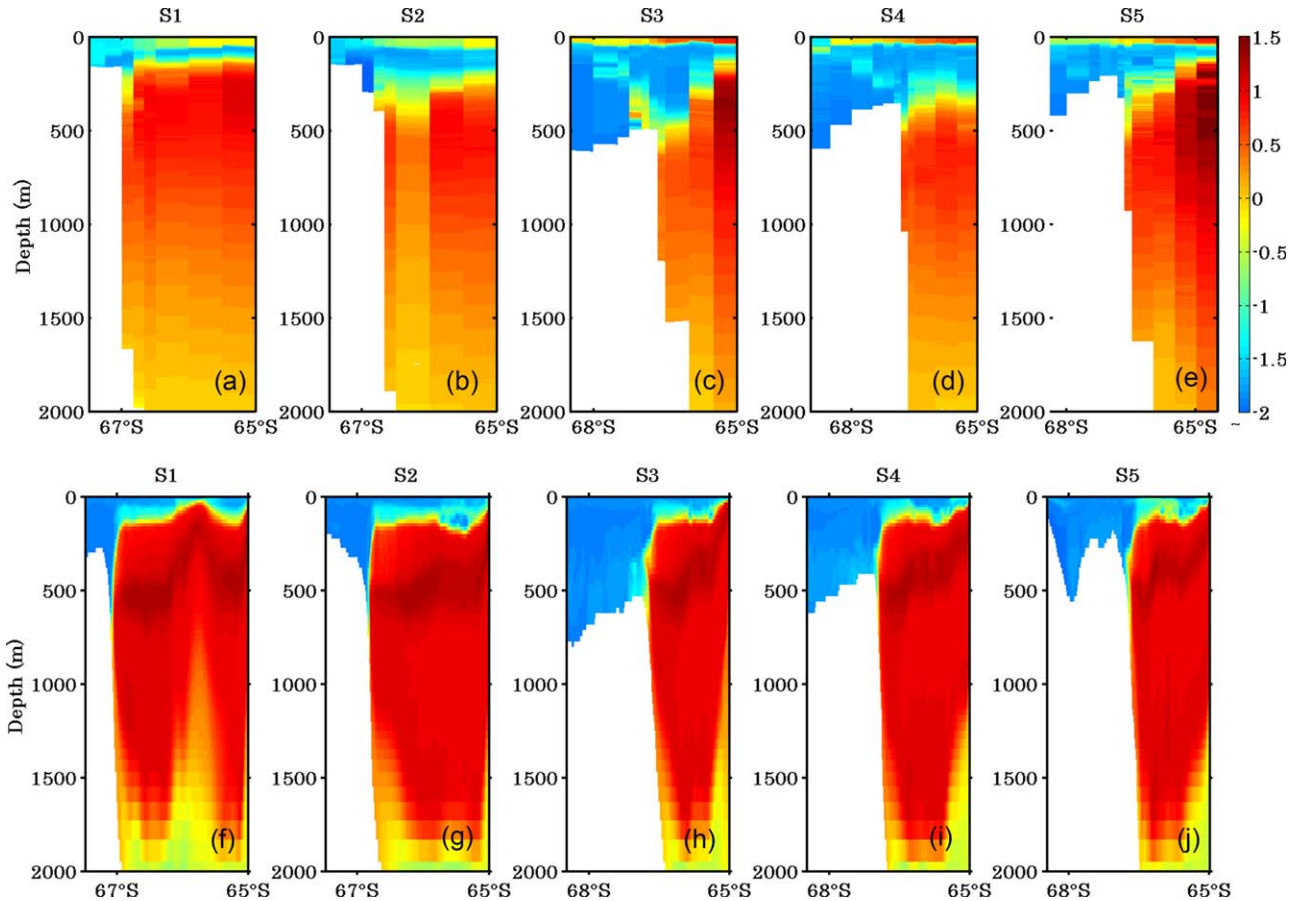

Figure 5. (top) Five potential temperature (°C) cross sections derived from the 29th CHINARE. S1-S5 denote the positions of five meridional sections (68°E, 70.5°E, 73°E, 75.5°E and 78°E) as marked in Figure 1a. (bottom) Same as the top plots except they are a snapshot of our simulation in summer.

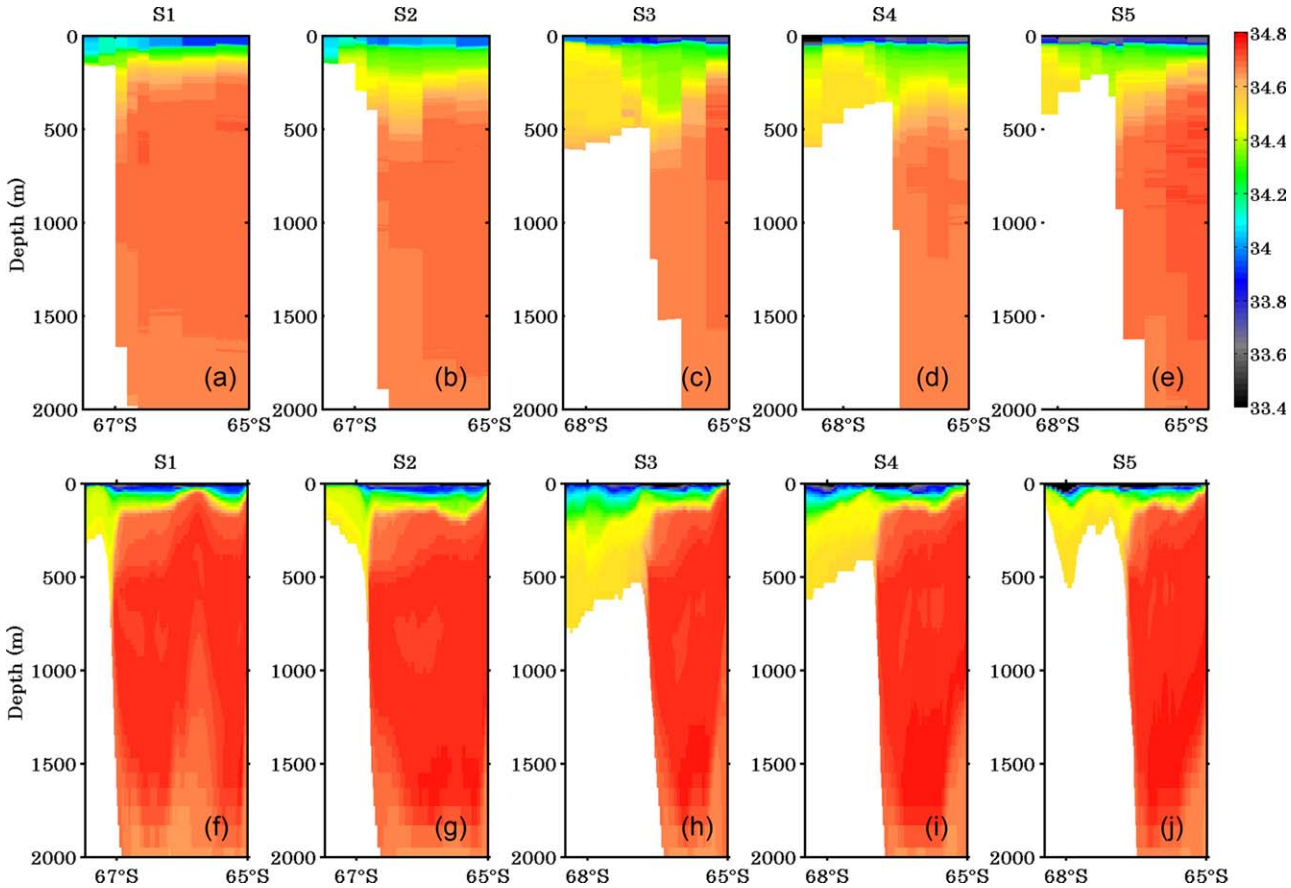

Figure 6. Same as Figure 5, but for salinity (psu).

north outside Prydz Bay, correlated with the density of the sea-ice coverage. Features of AASW are highly variable year by year due to large changes in local sea ice and atmospheric conditions [*Smith et al*., 1984]. The simulated AASW seems colder than hydrographic observation, which is related to the larger area of simulated sea-ice in summer. The subsurface WW layer almost occupies the entire layer between the seasonal thermocline and the permanent thermocline. WW could keep the memory of the previous winter, with a potential temperature close to the surface freezing point, and its thickness becomes thinner while extending offshore (Figure 5). Characterized as a warmer and saltier layer, the core of CDW is at a depth of ~500 m in the deep ocean, with a maximal potential temperature and salinity of ~1.5°C and ~34.65 psu (Figures 5 and 6). Evident southward intrusions of mCDW are observed in the 29th CHINARE, recognizable as onshore warm signals reaching as far south as 68°S at 73°E (Figure 5c) and 75.5°E (Figure 5d), which can also be seen in the result of *Williams et al*. [2016]. Although the onshore transport of mCDW is somehow weaker and deeper in our simulation, the simulated southward intrusions of warm signals are still captured at profiles of 73°E (Figure 5h) and 75.5°E (Figure 5i). The transition from CDW to mCDW occurs as soon as CDW mixes with colder SW, e.g., when CDW is trapped in the small troughs to the east of the FLB. Once being advected onto the continental shelf, warmer and saltier CDW mixes with colder and saltier SW and is transformed into mCDW. The simulated warm signals of mCDW are centered at ~400 m depth around 76°E (Figure 7c), with a maximal potential temperature of ~-1°C, slightly weaker and deeper than in JARPA (Figure 7a). In contrast to the salinity profile of JARPA (Figure 7b), a remarkable fresher type of water is identified between 70°E and 72.5°E in our simulation (Figure 7d). This fresher water is likely linked to the outflow of fresher ISW along the west coastline from the sub-ice-shelf cavity, suggesting that denser SW formation is hindered by the outflowing freshwater from beneath the AIS [*Williams et al*., 2016]. Compared with water masses observed at borehole sites (AM01-AM06, Figure 1a), the simulated ISW in the sub-ice-shelf cavity is largely consistent with these borehole-derived estimates (Figure 8), except for the sites of AM03 and AM04,

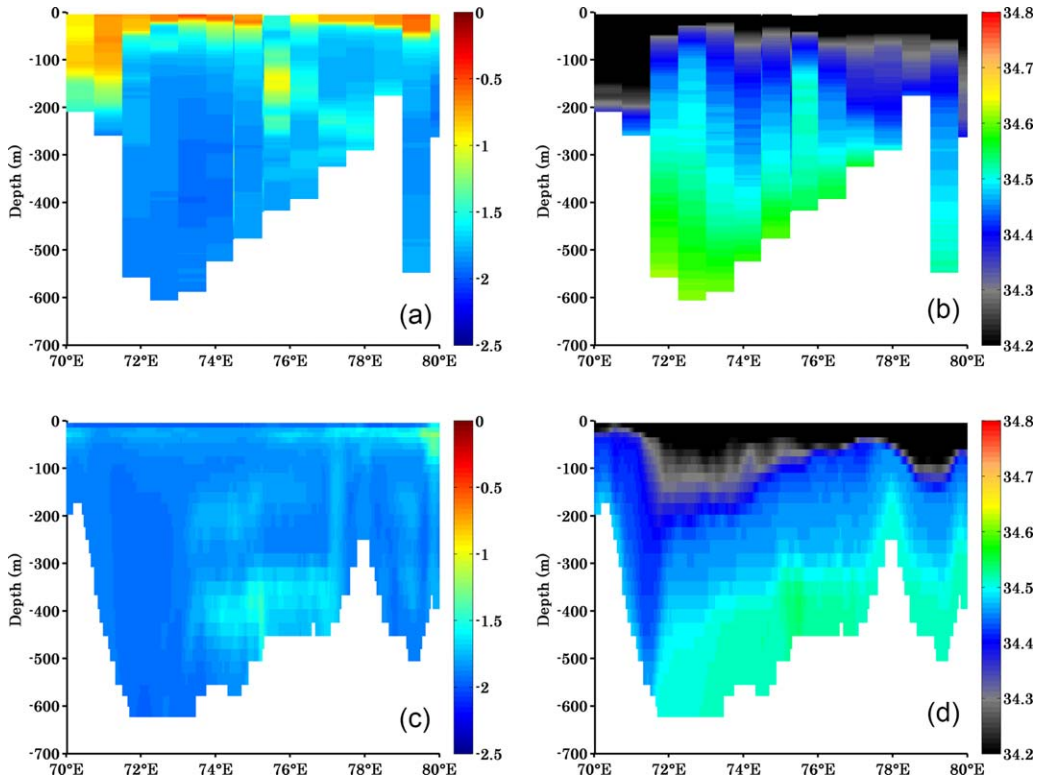

Figure 7. (a) Potential temperature (°C) and salinity (psu) cross sections at 67.5°S derived from the voyage of JARPA, as marked by red dots in Figure 1a. (c and d) Same as Figures 7a and 7b except they are the averages over summer from our simulation.

where the simulated ISW is somehow fresher than that observed. The differences between the simulation and observation in salinity of ISW may be caused by both the larger basal melting rate and the less production of high salinity Shelf Water (SW). Essentially, comparing to the hydrographic and borehole observations, our model captures the properties of water masses in summer and captures the mCDW intrusions across the continental shelf break.

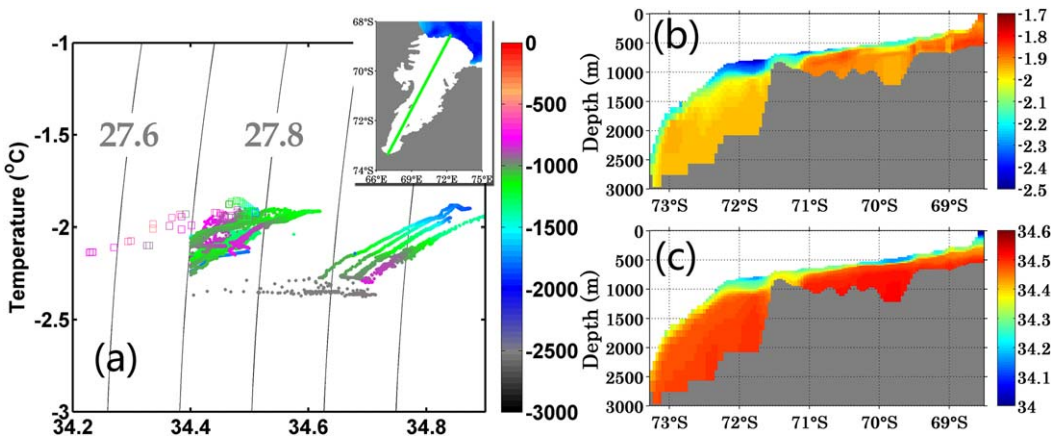

Figure 8. (a) Potential temperature (°C) and salinity (psu) diagram of HWD (closed circles, as marked by black squares in Figure 1a) and the averages over summer from our simulation (open squares) at sites of AM01 (69.44°S, 71.42°E), AM02 (69.71°S, 72.64°E), AM03 (70.55°S, 70.34°E), AM04 (69.9°S, 70.29°E), AM05 (70.23°S, 69.68°E), AM06 (70.24°S, 71.34°E), superimposed with potential density with contour intervals of 0.1 kg m^{-3} (black solid lines). The color scale denotes the depths in units of m. The green line in the inset in Figure 8a shows the position of a cross section shown in Figures 8b and 8c. (b) The cross section of potential temperature (°C) along the green line in the inset in Figure 8a. (c) Same as Figure 8b, but for the salinity (psu).

4. mCDW Intrusions and Transport

4.1. Locations for the mCDW Intrusions

The annual mean depth averaged velocity preliminarily shows the hot spots where mCDW intrudes onto the continental shelf (Figure 9). First, a strong longshore current, known as the ASC, is evidently identified in Figure 9a, consistent with observations from moorings and Argo float trajectories [*Heywood et al*., 1999; *Aoki et al*., 2010]. As the southward-shallower continental shelf break is dominated by the sharp ASF (Figures 5 and 6), the associated ASC features westward flow throughout the water column, even with subsurface or bottom intensification [*Heywood et al*., 1999]. In our simulation, the bottom intensified ASC across the 73°E section is characterized by a westward maximum of > -0.2 m s^{-1} at 66.5°S (Figure 9b), implying that strong onshore intrusions might be induced by the dynamic effects of topography on the ASC [*Klinck and Dinniman*, 2010; *St-Laurent et al*., 2013]. Second, there is a strong southwestward current along the coastline from 75°E to 82°E (Figure 9a). This current originates in a series of small troughs that are oriented perpendicular to the shelf break between 82°E and 90°E. After crossing the shelf break, this current flows southward by following the eastern side of the Four Ladies Bank (FLB) until it arrives at the coastline. Steered by the coastline, this current migrates southwestward as a strong coastal current. After passing the southwestern tip of the FLB, it finally merges with the eastern branch of the cyclonic PBG. Hereafter, we name this coastal current as the Prydz Bay Eastern Coastal Current (PBECC). Indeed, observation-based assessments of the geopotential height anomaly have shown a larger gradient to the south of the FLB [see *Smith et al*., 1984, Figure 10; *Middleton and Humphries*, 1989, Figure 2], suggesting the existence of the PBECC. Recently, observations using hydrographic profiles from seals have also revealed the presence of low dynamic height to the south of the FLB [*Herraiz-Borreguero et al*., 2015]. In addition, a modeling study with relatively lower (0.1° × 0.1°) spatial resolution also captures the PBECC as an along-coast current at 200 m depth [see *Aoki et al*., 2010, Figure 6]. Therefore, the simulation in this study, that has the highest horizontal resolution so far, along with the results in the above mentioned previous studies, evidently verifies the existence of the PBECC. This PBECC is likely to be a secondary current contributing to the mCDW

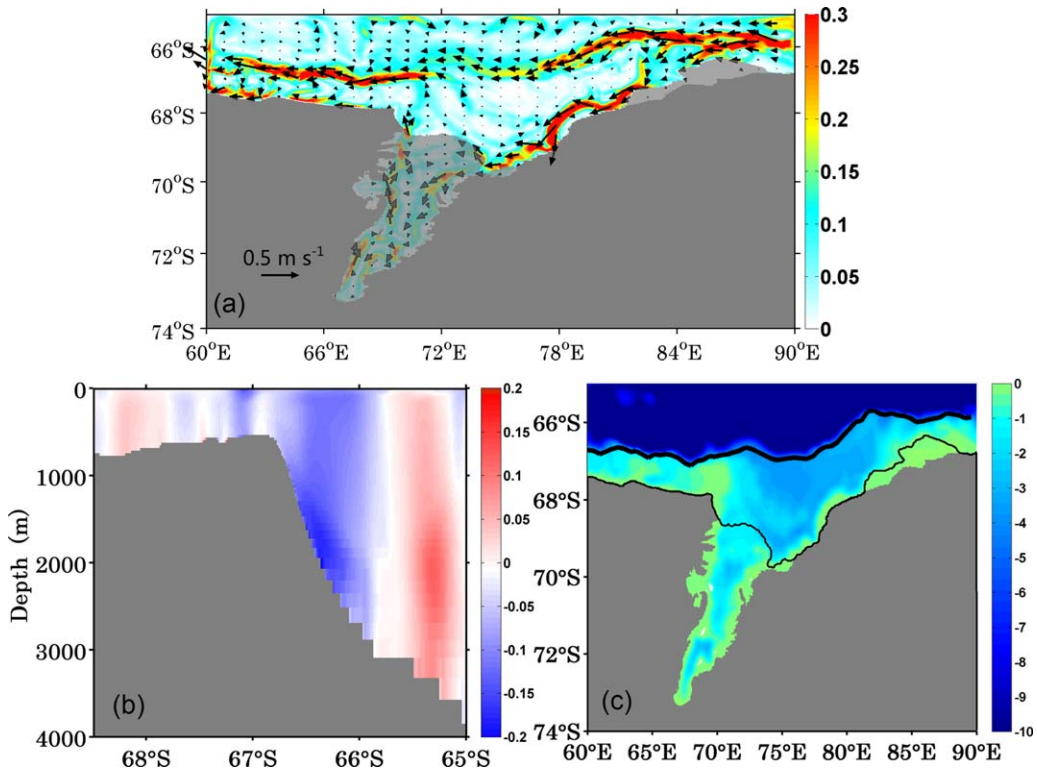

Figure 9. (a) The annual mean magnitude of depth averaged velocity (color shaded) in unit of m^1 s^{-1}, superimposed with the corresponding velocity vectors in units of m s^{-1}. The semitransparent region denotes the sub-ice-shelf cavity. (b) Zonal velocity cross section at 73°E in units of m s^{-1} (Eastward is positive). (c) The streamfunction obtained from vertically integrated zonal volume transport (Sv). The thick solid black line denotes the latitude of the 700 m isobath. The thin solid black line is the ice-shelf fronts and the coastlines.

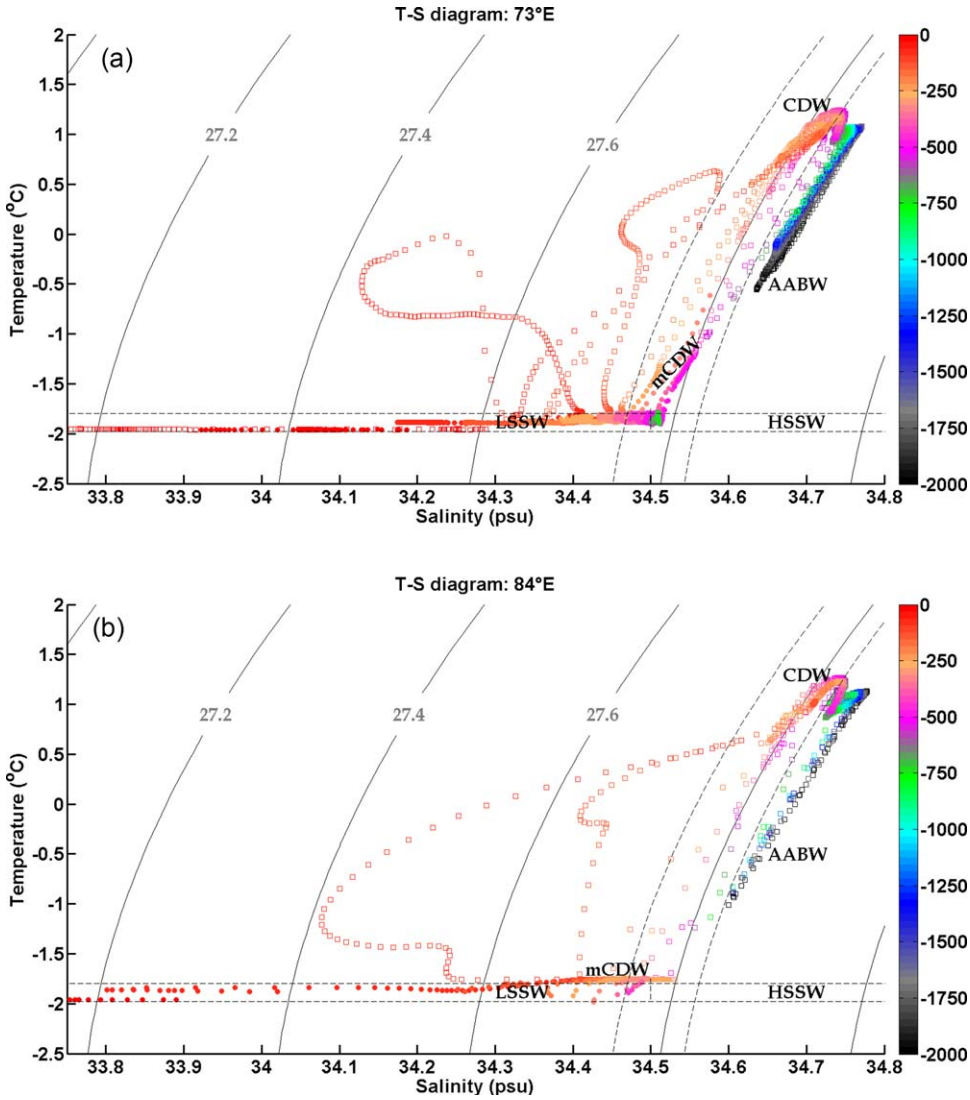

Figure 10. The average potential temperature (°C) and salinity (psu) diagram for the section of (a) 73°E and (b) 84°E in winter, superimposed with potential density in contour intervals of 0.2 kg m^{-3} (grey solid lines). Locations of water masses are represented by open squares (in the deep sea) and closed circles (on the continental shelf). The thin dashed lines define the water masses. The color bar indicates the depth of water column (m). Acronyms are defined in the text, except High Salinity Shelf Water (HSSW) and Low Salinity Shelf Water (LSSW).

intrusions in addition to the PBG, as hypothesized by *Herraiz-Borreguero et al*. [2015]. The mechanism responsible for the PBECC maintenance could be primarily explained by the restriction of potential vorticity. In order to ensure conservation of potential vorticity, after crossing the continental shelf break at troughs, the upstream PBECC is likely to flow along the isobaths of ~300 m. As a result, the PBECC is steered by the steep topography along the coastline, due to the presence of the FLB where the seabed is shallower than 200 m (Figure 1a). The barotropic streamfunction obtained from vertically integrated zonal volume transport shows that the cross shelf exchanges mostly occurs to the east of 73°E, and the onshore mCDW intrusions occur in both the eastern branch of the PBG and the upstream of the PBECC perpendicular to the continental shelf break (Figure 9c).

We chose two representative meridional cross sections at 73°E and 84°E to reveal the water mass classifications at the Amery Depression and the troughs to the east of the FLB, respectively (Figure 10). At the northern tip of the Amery Depression (73°E) where warmer CDW (~1°C, ~34.7 psu) to the south of the continental shelf break is advected southeastward at depths of about 250–500 m, CDW gradually mixes with cooler and fresher SW and is transformed into mCDW (~-1.5°C, ~34.55 psu) on the continental shelf

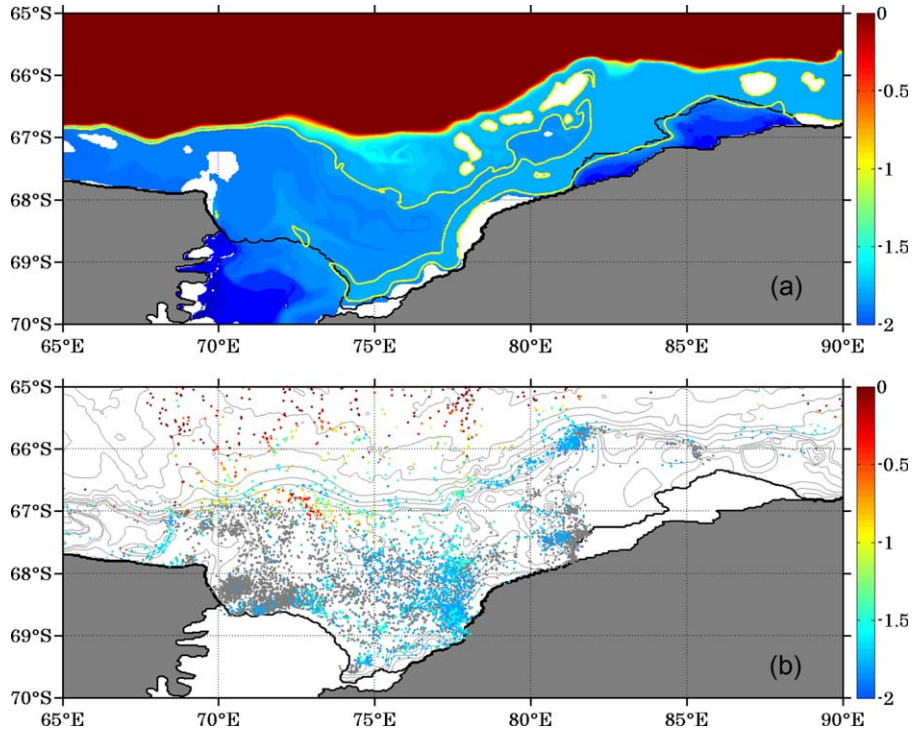

Figure 11. (a) The depth averaged (between 300 m and 500 m) potential temperature in winter (°C). Yellow lines are at −1.8°C. (b) Spatial distribution of the potential temperature (°C) retrieved by instrumented southern elephant seals at depths > 300 m during 2009, 2011, 2012, and 2015. Grey dots show the profiles with potential temperature < −1.8°C. Grey solid lines are contours of bottom depth (at 100 m intervals from 100 to 500 m, and 500 m intervals thereafter). Black dashed lines are the ice-shelf fronts and the coastlines.

(Figure 10a). In contrast to the section at 73°E, the transformation of CDW at the section at 84°E is much more intense, leading to the formation of mCDW, with a potential temperature of ∼-1.7°C and salinity of ∼34.45 psu (Figure 10b). Interestingly, the characteristics of this mCDW formed at 84°E are very consistent with the mCDW observed by moorings on the eastern flank of the AIS calving front in winter [*Herraiz-Borreguero et al*., 2015, 2016], implying that one of the sources of mCDW observed near the AIS may be located at the small troughs to the east of the FLB. Based on the assessments of the depth-averaged temperature between 250 m and 500 m in winter (Figure 11a), two critical windows have been clearly identified as the locations where strong onshore heat transport occurs. The first one, documented in previous studies [*Smith et al*., 1984; *Nunes Vaz and Lennon*, 1996; *Galton-Fenzi et al*., 2012], is located at the northwestern tip of the Amery Depression. Along the Amery Depression, mCDW rides above the inflow of the PBG and hugs the western rim of the FLB. The second one, highlighted in this study, is located at the small troughs to the east of the FLB where mCDW is formed and advected downstream to the eastern flank of the AIS calving front (Figure 11a). The spatial distribution of temperature collected by instrumented southern elephant seals also confirms the existence of the mCDW to the east and the south of the FLB, although seals have not frequently visited these troughs (Figure 11b). Therefore, both the PBG and the PBECC serve to transport mCDW onto the continental shelf in compensation for the outflow of SW, confirming the inference of *Herraiz-Borreguero et al*. [2015].

4.2. Seasonality of Circulations and Onshore Heat Transport

For convenience, we chose two meridional sections crossing 75°E (marked by thick red and green lines in Figure 12a) to diagnose the transport of the ASC and the PBECC. The transports of the ASC and the PBECC peak in June, with a maximum of ∼29.3 Sv and ∼3.8 Sv, respectively (Figure 12b). After June, both the ASC and the PBECC gradually get a slowdown until summer, decreasing to ∼15.6 Sv and ∼2.2 Sv, respectively. The seasonal cycle of the ASC agrees well with the estimations from moorings [*Heywood et al*., 1999], and coincides with the seasonality of the PBECC. Essentially, the transport of the PBECC can be considered as an indicator of the influx at upstream troughs, reflecting that the seasonal pulsing of the mCDW intrusions is dependent on the strength of the ASC. This simulated temporal coherency between the strength of ASC

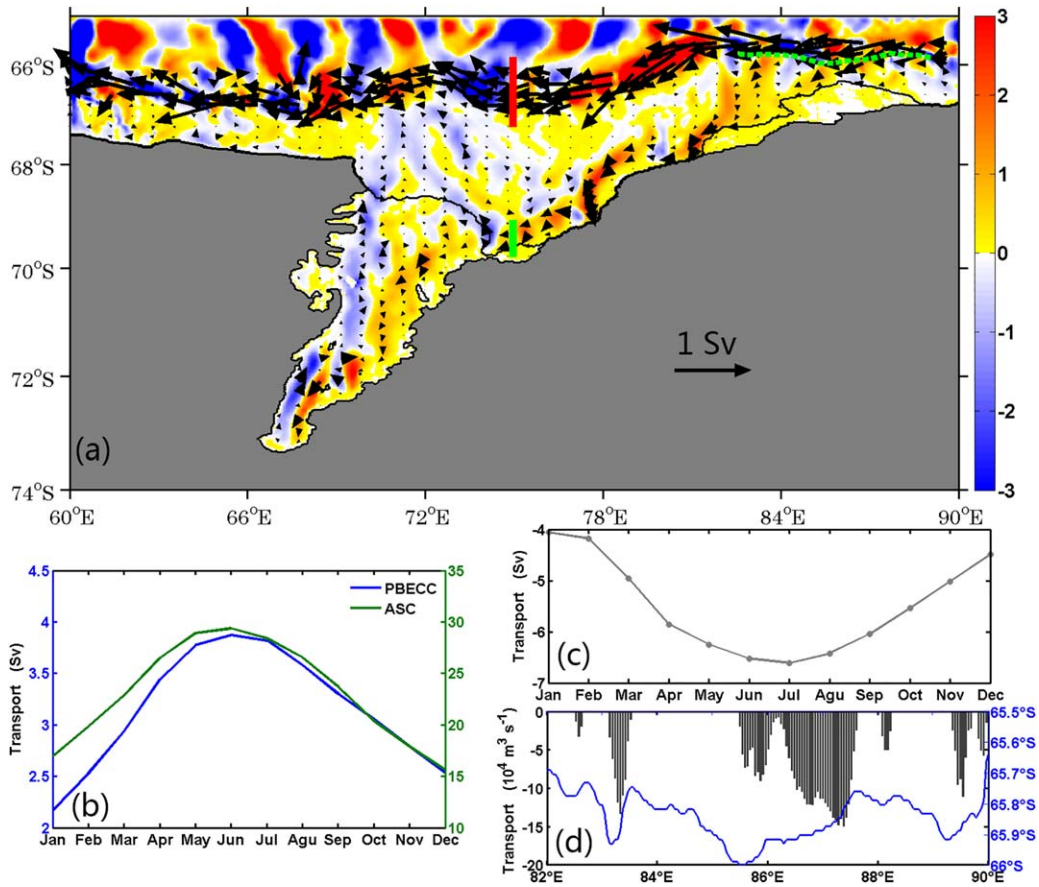

Figure 12. (a) Vectors are the annual mean vertically integrated transport (Sv). The vertical integrated meridional volume transport in color, in units of 10^{-1} Sv. (b) The monthly averaged transport of the ASC across the thick red line in the top panel and PBECC across the thick green line in the top panel at 75°E (Sv). (c) The zonally integrated onshore transport (across the isobaths of 700 m, the thick green dashed line in the top panel from 82°E to 90°E) to the east of the FLB, in units of Sv. (d) Black bars are the annual mean onshore transport across the latitudes of 700 m isobaths to the east of the FLB (the green dashed line in the top plot, from 82°E to 90°E), in units of 10^4 m^3 s^{-1}. The blue line denotes the latitudes of 700 m isobaths along the continental shelf break.

and the mCDW intrusions agrees well with previous idealized model results [*Klinck and Dinniman*, 2010; *St-Laurent et al*., 2013]. In order to assess the cross-shelf exchanges to the east of the FLB, we first calculated the onshore volume transport between 82°E and 90°E. As expected, the monthly integrated onshore transport reaches its peak value of ~6.6 Sv in July (Figure 12c, note that negative values mean onshore transport), implying that the intrusions of mCDW onto the continental shelf are mostly confined to the wintertime. The latitudes of 700 m isobaths and the corresponding annual mean meridional heat transport between 82°E and 90°E clearly shows the spatial correlations between the cross-shelf heat transport and the locations of troughs (Figure 12d, note that negative values mean onshore transport). In *St-Laurent et al*. [2013], it has been found that the onshore intrusions mainly occur along the east side of troughs, as a consequence of potential vorticity conservation. The zonally integrated onshore volume transport between 82°E and 90°E could be up to ~5.5 Sv, thus exerting great impacts on downstream water masses and the basal mass balance of the AIS.

We further assessed the eddy kinetic energy and onshore heat transport to diagnose the contribution of meso-scale processes. The eddy kinetic energy reveals that the ASC hosts energetic meso-scale eddies, especially in the upstream to the northeast of the FLB (Figure 13a). In order to evaluate the respective contribution of eddies and mean flow to the onshore heat transport, we calculated the oceanic heat transport Q at any given time t using the following definition [*Volkov et al*., 2008]:

$$Q(t)=\int\int \rho C_p T v dz dx, \tag{2}$$

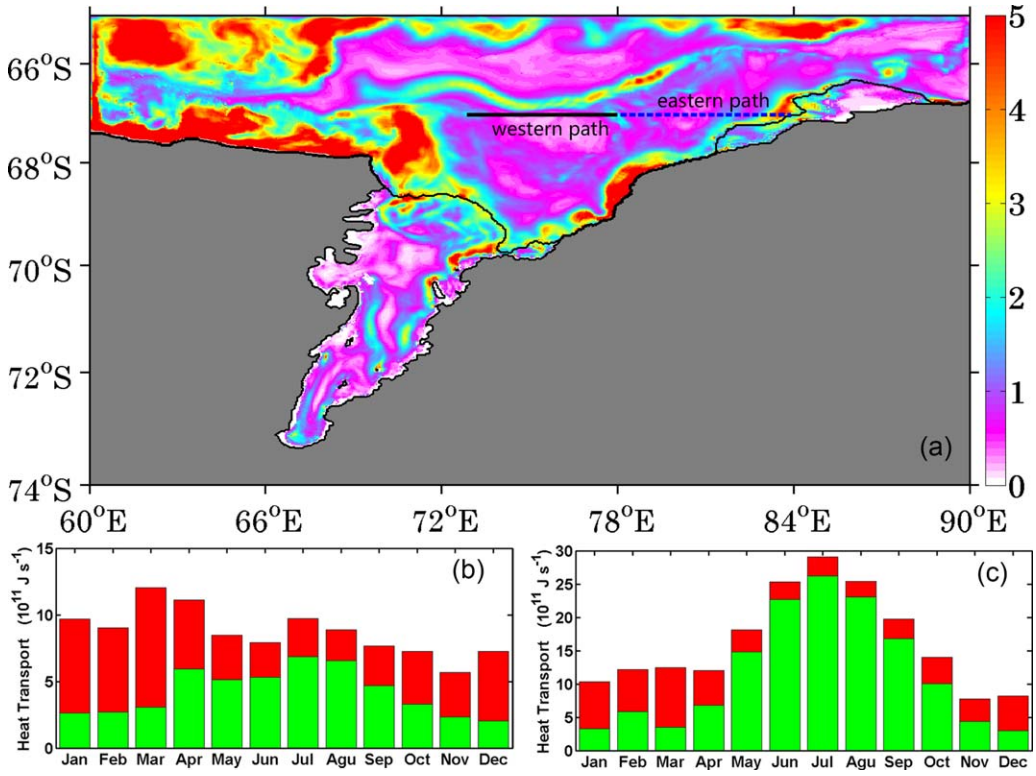

Figure 13. (a) The eddy kinetic energy in color, in units of 10^{-4} m^2 s^{-2}. (b) The zonally integrated onshore heat transport across 67.5°S (the thick black solid line in the top plot from 73°E to 78°E). The total onshore heat transport has been decomposed into contributions from the mean flow (green) and eddies (red), in units of 10^{11} J s^{-1}. (c) Same as Figure 13b, but for the onshore heat transport across the thick blue dashed line in the top plot (from 78°E to 83°E).

$$T=\theta-\theta_f, \tag{3}$$

where ρ is the density of seawater, C_p is the specific heat capacity, T is the deviation of the potential temperature θ from the surface freezing point θ_f and v is the meridional velocity. Thus, the oceanic meridional eddy heat transport Q_e can be estimated by the deviations from the time-mean value:

$$Q_e(t)=\int\int \rho C_p Tv dzdx-\int\int \rho C_p \overline{Tv} dzdx, \tag{4}$$

where the overbar indicates averaging over a 1 month interval. For convenience, we chose two zonal sections to diagnose the southward heat transport along these two paths (marked by the thick black solid line and the thick blue dashed line in Figure 13a, respectively). For both eastern and western paths, the contribution of mean-flow plays an important role in the onshore heat transport, especially in modulating the seasonality of onshore intrusions. However, the eddy-induced heat transport accounts for ~23% in the eastern path and ~52% in the western path with respect to their total onshore heat transport, respectively. This result thus indicates that the eddy-induced onshore heat transport has a greater impact in the western path (Figure 13b) than in the eastern path, although its absolute magnitude is relatively smaller than that to the east of the FLB (Figure 13c).

4.3. Intrusions of mCDW into the AIS Cavity in Winter

As previous studies have highlighted that mCDW observed by seal profiles arrives at the eastern flank of the AIS calving front at the beginning of the austral winter [*Herraiz-Borreguero et al.*, 2015, 2016], we further assessed the mCDW intrusions into the sub-ice-shelf cavity, and melting/freezing rate thereby induced, of the AIS in winter. Melting and freezing rates at the base the ice shelf are primarily determined by the differences between the temperature of seawater in contact with the ice shelf and the *in situ* freezing point temperature. Here, the *in situ* freezing point temperature of seawater T_f is defined from a linearized function of salinity and pressure [*Galton-Fenzi et al.*, 2012]:

$$T_f = aS + b + cP, \tag{5}$$

where a is -5.73×10^{-2}°C psu^{-1}, b is 8.32×10^{-2}°C, c is -7.61×10^{-2}°C dbar^{-1}, S is the salinity (psu) and P is the pressure (dbar).

Generally, freezing occurs along the west and north of the AIS, and melting occupies the east and south of the AIS (Figure 14a), consistent with previous studies [*Galton-Fenzi et al.*, 2012; *Herraiz-Borreguero et al.*, 2015]. The basal mass balance reflects the ocean thermal forcing at the ice-shelf-ocean interface, that is principally dominated by the potential temperature of seawater in contact with the ice shelf (Figure 14b) and the circulation pattern (Figure 14d). In the eastern flank, the inflow of mCDW into the AIS cavity appears as eddy-like flows [*Herraiz-Borreguero et al.*, 2016], and becomes cooler when descending southward (Figure 14b). The southernmost ISW is formed by the mixing of the inflow water with local cold and melted fresher water, driving a maximal melt rate of ~30 m yr^{-1} at the grounding line (Figure 14a). The high melt rate at the grounding line agrees with estimations from previous studies [*Wen et al.*, 2007; *Galton-Fenzi et al.*, 2012]. In the western flank, the outflow of ISW ascends under the AIS and gradually becomes super-cooled (cooler than the *in situ* freezing point temperature, Figure 14c), leading to areas of freezing (Figure 14a). The seasonality of simulated basal mass balance also agrees well with previous modeling studies (not shown).

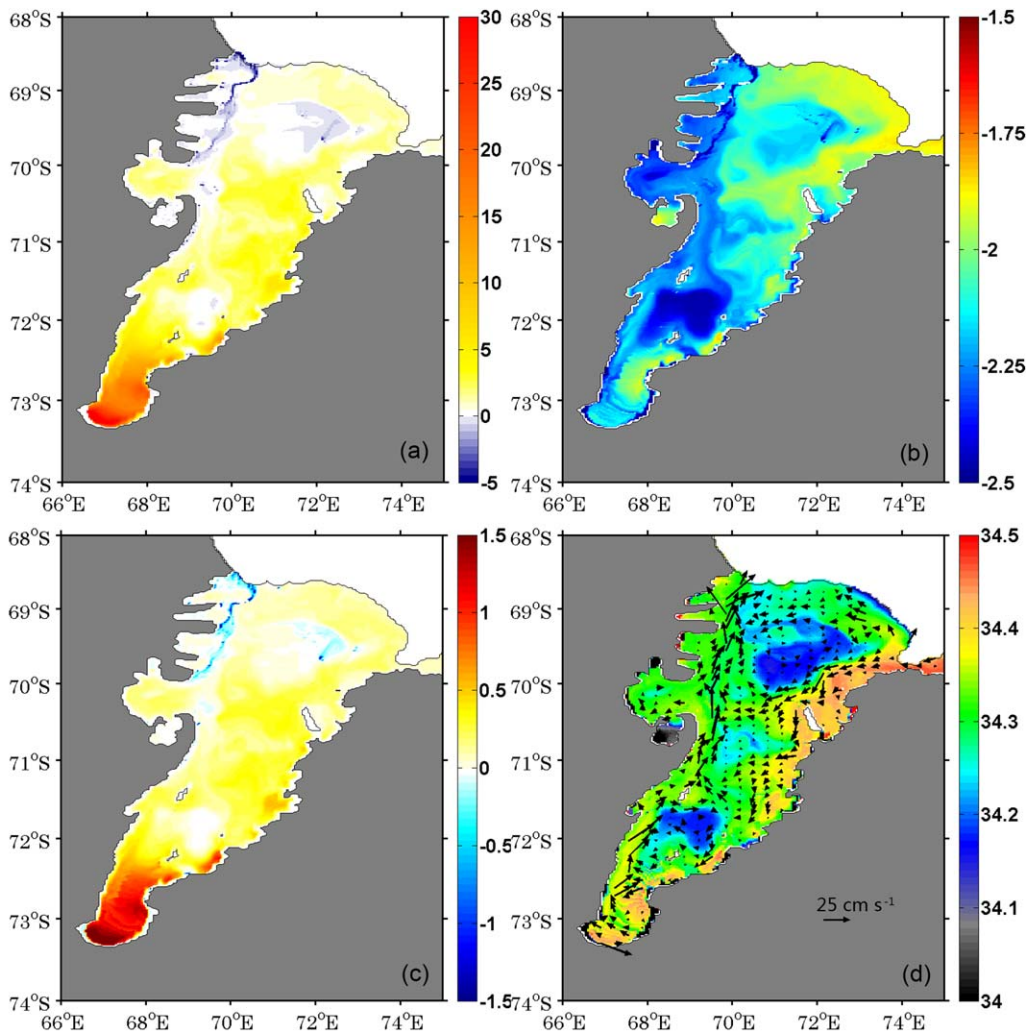

Figure 14. (a) The fresh water flux (freezing is negative and melting is positive) at the ice shelf-ocean interface of the AIS in winter, in units of m year^{-1}. (b) The temperature (°C) of the layer adjacent to the AIS. (c) The differences between the temperature of the layer adjacent to the ice shelf and the local freezing point temperature (°C). (d) The salinity (psu) of the layer adjacent to the AIS, superimposed by corresponding currents (cm s^{-1}). Black lines are the margin of the AIS.

5. Conclusions

In this paper, we identified the locations where mCDW intrudes onto the continental shelf and the flow pathways through which mCDW is delivered to the AIS calving front, using a eddy-resolving regional coupled ocean-sea ice-ice shelf circulation model for Prydz Bay and the AIS. The model results are generally consistent with observations derived from satellite, hydrographic and borehole data, though future work is necessary for better simulating the coupled ocean-sea ice-ice shelf system in Prydz Bay.

In contrast to previous modeling studies of Prydz Bay, a longshore flow along the continental shelf break, i.e., the westward ASC, is resolved in our simulation. The dynamic effects of the continental shelf break on the ASC have great potential for producing onshore heat transport. Two critical regions where mCDW effectively intrudes onto the continental shelf are identified. The first one, well documented in previous studies [*Smith et al*., 1984; *Nunes Vaz and Lennon*, 1996; *Yabuki et al*., 2006; *Galton-Fenzi et al*., 2012], is the eastern branch of the PBG along the Amery Depression. The second one, highlighted in this study as the PBECC, originates in a series of small troughs to the east of the FLB. Both the PBG and the PBECC serve to transport mCDW to the AIS calving front with potential consequences for the basal mass balance of the AIS. The annual mean volume transport is estimated at ~0.58 Sv by the western path and ~3.24 Sv by the eastern path in Figure 13, with the annual mean heat transport of ~8.7 $\times 10^{11}$ J s^{-1} and ~16.2 $\times 10^{11}$ J s^{-1}, respectively. Note that the mCDW carried by the PBECC is colder than that carried by the PBG. Figure 15 shows the schematic diagram of primary circulations in Prydz Bay. As a few small branches split from the ASC to the east of the FLB, the upstream PBECC originates in the small troughs perpendicular to the shelf break. Since the seabed over the FLB is around 200 m deep, the water masses in the eastern bay (~500 m depths) cannot flow westward directly due to the constraint of topography. To keep the conservation of potential vorticity, the water masses in the eastern coastal region are squeezed into a narrow deeper path (~500 m) that is closely adjacent to the eastern coast, and the upstream PBECC is deformed into a strong jet encircling the northeastern side of the FLB. After passing the southwestern edge of the FLB, the downstream PBECC merges with the eastern branch of the PBG. Our results show that both the PBG and the PBECC serve to transport mCDW to the AIS calving front, well consistent with the inference of *Herraiz-Borreguero et al*. [2015]. After mCDW arrives at the eastern flank of the AIS calving front, it descends and flows southward beneath the AIS and mixes with fresher melt-water, leading to enhanced ice-shelf basal melting in winter. When this inflow encounters variations of the thickness of water columns beneath the AIS, four clockwise recirculation features are produced (Figure 14c). In the western flank, the outflow of ISW ascends and flows northward and gradually becomes super-cooled, giving rise to refreezing in the northwestern part of the AIS. On the western flank of the AIS calving front, the diluting factor of outflowing ISW could effectively suppress the impact of brine rejection on the formation of denser SW in the MP [*Williams et al*., 2016]. The majority of this relatively fresher SW flows along the western boundary of Prydz Bay out toward the continental shelf break, with some recirculation back toward the AIS. After flowing directly out of Prydz Bay, this

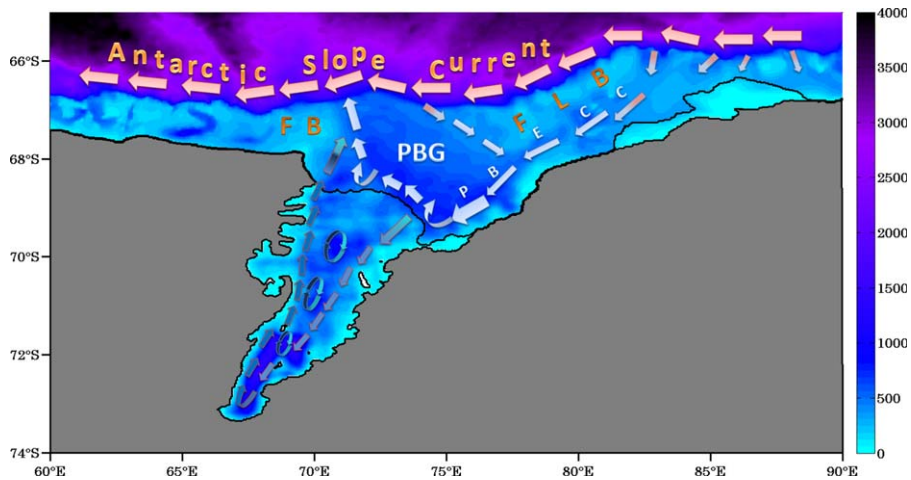

Figure 15. Schematic diagram of the flow pattern in Prydz Bay. The color scale shows the water column thickness in unit of m. The FB and FLB are marked at their corresponding locations.

fresher SW primarily migrates northwestward on the continental slope and merges with the westward ASC (Figure 15).

This work attempts to investigate the importance of higher spatial resolution and more accurate topography data in modeling the dynamic effects of the continental shelf break on the ASC. If the spatial resolution was limited to only resolving relatively large bathymetric irregularities along the shelf break, such as the depressions around Antarctica, the dynamic effects of small-scale troughs and associated cross-shelf exchanges could not be captured. It is thus important to resolve the dynamic effects of small troughs when developing numerical models for investigating cross-shelf exchanges and associated onshore mCDW transport around Antarctica. Note that we did not conduct an experiment with coarse spatial resolution to further confirm the influence of higher resolution. Nevertheless, a number of previous idealized or realistic model studies have been focused on the influence of model resolution and bathymetry accuracy [e.g., *Klinck and Dinniman*, 2010; *St-Laurent et al*., 2013]. All of these studies strongly stressed the key role of spatial resolution in modeling the dynamic effects of topography on the ASC.

Although profiles retrieved by instrumented southern elephant seals have been used to describe the distribution of mCDW in Prydz Bay, hydrographic observations are still too sparse to give a full picture of the delivery of mCDW to the inner continental shelf. We suggest deploying moorings to monitor the heat transport along the path of the PBECC, and then quantitatively distinguishing the respective contribution of the PBG and the PBECC to the onshore heat transport.

We note that mCDW intrusions induced by the eastern branch of the PBG seem to be weaker in our simulation than in previous studies [*Galton-Fenzi et al*., 2012; *Williams et al*., 2016]. This apparently results from a relatively weaker PBG (Figure 12a). The largest uncertainty of our simulation arises from open boundary conditions. The ASC simulated in ECCO2 is relatively weaker than the mooring-based observation in *Heywood et al*. [1999]. Although the observed ASC applied at the open boundaries could improve the simulated ASC, the short time duration of observation still cannot capture the annual variability in the ASC. It is necessary to assess the strength of the ASC with different representations in multiple reanalysis date sets, e.g., the most recently released ECCO-v4 [*Forget et al*., 2015]. Meanwhile, the simulated warmer and saltier CDW in the deep ocean may also result from the biases in the temperature and salinity on the open boundaries (Figures 5 and 6). Based on the model validation near the continental shelf break, the strong warmer biases in the simulated CDW has the potential to make us overestimate the onshore heat transport by increasing the temperature of mCDW intrusion, especially in the path of the PBECC (the eastern path in Figure 13a). The lack of tidal processes in our simulation may also lead to weaker mCDW intrusions [*Galton-Fenzi et al*., 2012], which might be the reasons for colder cross-shelf mCDW in simulations than in hydrographic observations (Figure 5). As a result, the simulated mCDW intrusions may be improved by a better representation of onshore cross-shelf warm signals, implying that our simulation might still underestimated onshore heat transport in Prydz Bay, especially in the eastern branch of the PBG.

The uncertainties of SW properties may also be induced by the biases in the atmospheric forcing conditions. Previous studies have identified large biases in atmospheric reanalysis data around Antarctica [*Wang et al*., 2016]. Specifically for the study region of this paper, we found that there are large uncertainties by comparing the results from reanalysis datasets with observations from weather stations within Prydz Bay. The biases in these surface boundary forcing could induce biases in sea ice coverage and movement and the extent of polynyas, and thereby result in biases in the properties of SW, e.g., the weaker polynyas in our simulation may result in less production of high salinity SW by weaker brine rejection. Thus, the simulated fresher ISW than borehole observations in the eastern blank of the sub-ice-shelf cavity (Figure 8) may partly result from the less production of high salinity SW over the PBP.

The bedrock and ice shelf thickness employed in our model domain is derived from an interpolation of RTopo-1 with a resolution of 1 min ($\sim$1.8 km) in zonal and meridional direction. In contrast to the International Bathymetric Chart of the Southern Ocean (IBSCO), the relatively lower resolution of RTopo-1 has the potential to miss some important topography features. The IBCSO is mapped with a resolution of 0.5 km, and thereby has a better representation of the smaller bathymetry features in Prydz Bay. However, RTopo-1 advances with a consistent representation of the shape of Antarctic sub-ice-shelf cavity and bedrock bathymetry. To preliminarily investigate the process in the sub-ice-shelf cavity and make a comparison with previous model studies [e.g., *Galton-Fenzi et al*., 2012], we chose RTopo-1 as the backbone of bathymetry in

this study, yet it also implies that our simulation still might underestimate the dynamic effect of topography on the ASC. In order to further assess the uncertainties induced by the biases in different bathymetry data sets, it is necessary to revisit the topography dynamic effects on the ASC with different topography data sets, e.g., a blend of IBCSO and RTopo-1.

The vertical resolution in a z coordinate model is crucial in representing the shape of the sub-ice-shelf cavity. In this study, we chose a compromise design (70 levels) based on the sensitivity experiments (50–100 levels) conducted by *Schodlok et al*. [2016], implying that our simulation could overestimate the basal melting rate of the AIS and yield the fresher simulated ISW than observations from borehole sites. Increasing vertical resolution in MITgcm could improve the simulated interactions at the ice-ocean interface beneath the ice shelf in Prydz Bay. It would be interesting to assess the dependence of the AIS melting rate on the vertical levels in our following studies, especially on the increasing vertical resolution near the depths of the grounding line.

We also note that our simulation has not included the frazil ice processes that have already been found to be important in determining the basal melting/freezing pattern of the AIS [*Galton-Fenzi et al*., 2012; *Herraiz-Borreguero et al*., 2013]. The omission of frazil ice processes is probably the cause of the relatively lower freezing rate at the base of the AIS than in *Galton-Fenzi et al*. [2012] and higher simulated sea-ice coverage than observed. To get a better understanding of the interactions between the ocean and the ice shelf in Prydz Bay, the inclusions of frazil ice processes in an eddy-resolving coupled regional ocean model will be a desirable work in the future.

Based on a high resolution ($1/10° \times 1/30°$) tidal model, *Hemer et al*. [2006] proposed that barotropic tidal processes had little influence on the oceanographic properties of the AIS cavity. The tidal current observed by moorings is also relatively weak (<0.05 cm s^{-1}) in Prydz Bay [*Ohshima et al*., 2013; *Nakayama et al*., 2014]. On the other hand, *Galton-Fenzi et al*. [2012] highlighted the effects of tidal processes under the AIS. It will be interesting to use an eddy-resolving model with frazil ice processes to revisit the tidal role in the basal mass balance of the AIS.

Acknowledgments
This work is supported by the Major State Basic Research Development Program of China (2016YFA0601804), by the China National Natural Science Foundation (NSFC) Project (41276200; 41306208; 41406214; 41376190; 41606217), by the Fundamental Research Funds for the Central Universities (2017B04814), by Ocean Public Welfare Scientific Research Project(201405031), by the Chinese Polar Environment Comprehensive Investigation & Assessment Program (CHINARE-2016-01-01), by the Special Program for China Meteorology Trade (grant GYHY201306020), by Program for Innovation Research and Entrepreneurship team in Jiangsu Province, by the Jiangsu Government Scholarship for Overseas Studies, by the China Scholarship Council (201504180026; 201708320046), and by a project funded by the Priority Academic Program Development of Jiangsu Higher Education Institutions (PAPD). Data from the 29th CHINARE are available through the Chinese National Arctic and Antarctic Data Center (http://www.chinare.org.cn/en/difDetailPublicDataset/?id=6725), and it is also available upon request to the corresponding author (Zhaomin Wang, zhaomin.wang@hhu.edu.cn). Hydrographic data from JARPA are available through the Institute of Cetacean Research (http://www.icrwhale.org/oceanograhicdata.html). Data from the boreholes are available through the Australia Antarctic Data Center (https://data.aad.gov.au/metadata/records/ASAC_1164). The instrumented southern elephant seal data are provided by the Integrated Marine Observing System (https://imos.aodn.org.au/imos123/home). The sea ice concentration data are provided by the National Snow and Ice Data Center (http://nsidc.org/data/NSIDC-0051#). Thanks to Benjamin Keith Galton-Fenzi (University of Tasmania, Australia) for advice on topography data, Dimitris Menemenlis (JPL, USA) for advice on using the MITgcm, and Andrew L. Stewart (UCLA, USA) for advice choosing on the momentum mixing parameterization scheme. We also thank Eileen Hofmann, John Klinck, Michael Dinniman and Pierre St-Laurent (CCPO, USA) for their constructive comments and discussions. The constructive comments of the two anonymous reviewers were greatly appreciated.

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